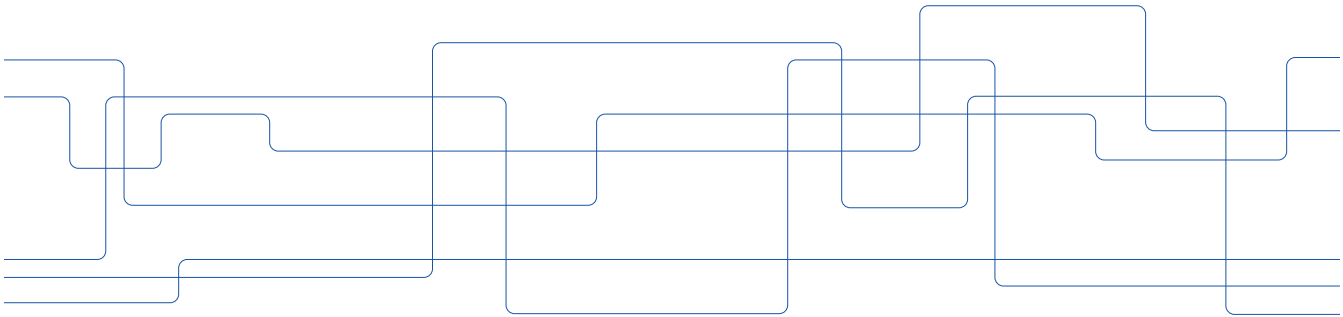
Molecular Simulations

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PDC Center for High Performance Computing

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Outline – Day 2

- Refresh on simulation methods
- Molecular Models
- What governs a simulation
- Set a simulation

One goal of a molecular simulation

to generate enough representative conformations of the molecular system in such a way that accurate values of a property can be obtained.

Which methods for molecular simulations?

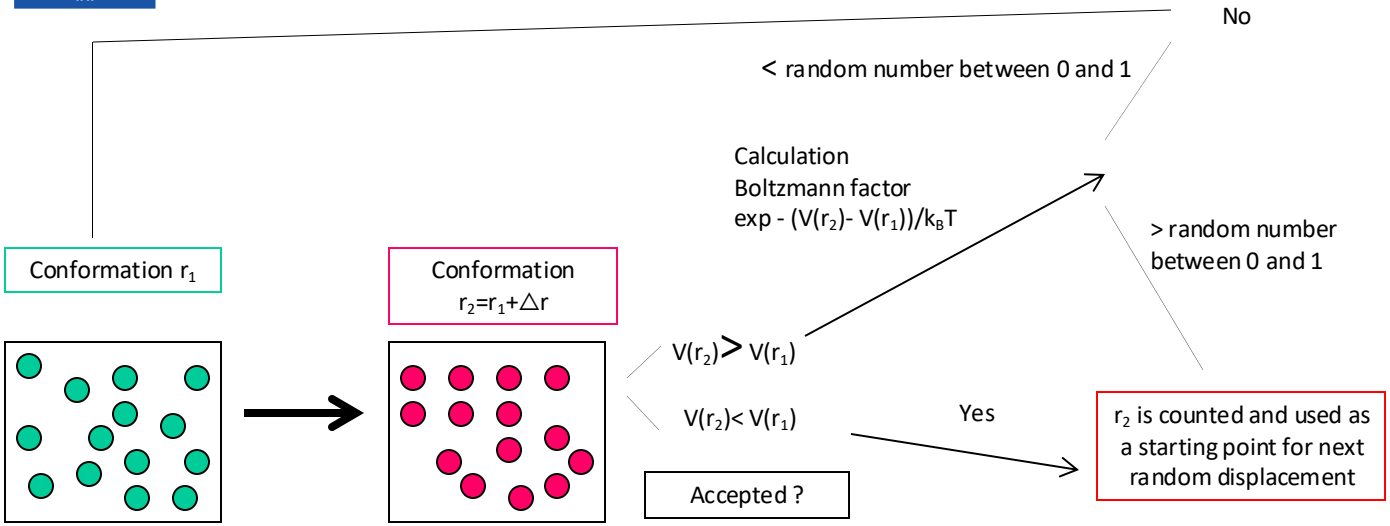
Monte-Carlo (MC) method

generates conformations by randomly generating new coordinates and energy criteria

Molecular Dynamics (MD) method

generates conformations by integrating Newton's law of motion

Monte Carlo simulation



Generation of new configuration by randomly moving an atom or several atoms

Calculation of energy of new configuration

Which methods for molecular simulations?

Monte-Carlo (MC) method

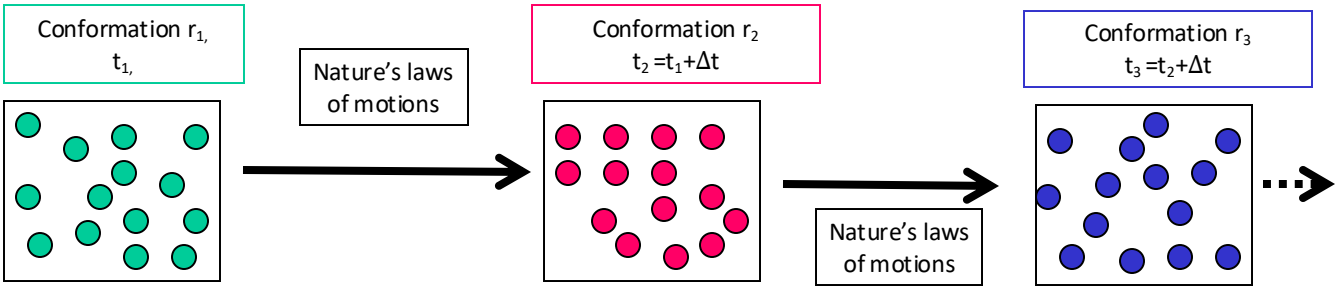
generates conformations by randomly generating new coordinates and energy criteria

Molecular Dynamics (MD) method

generates conformations by integrating Newton's law of motion

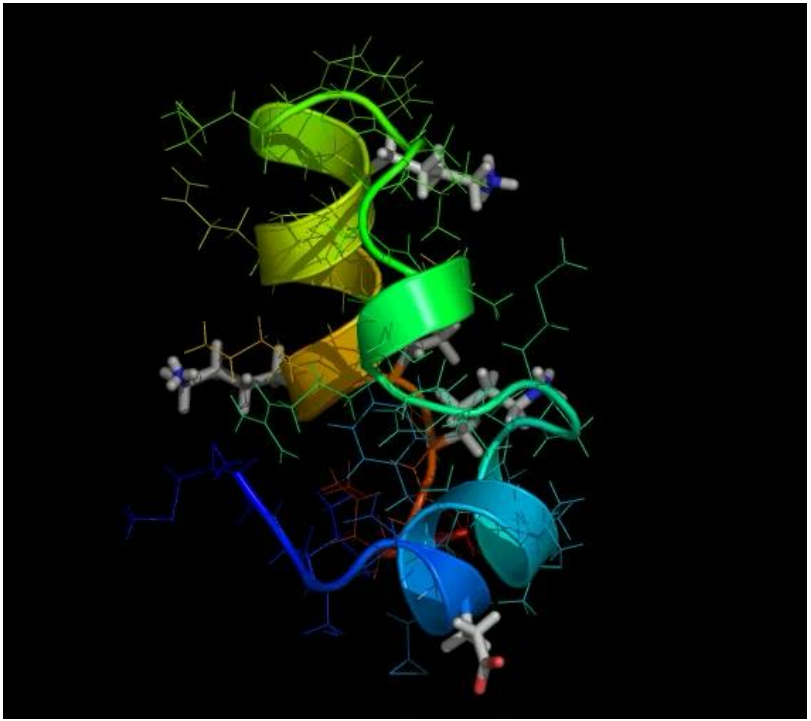
Molecular dynamics simulation

The Molecular Dynamics method generates configurations applying Nature's law of motion for atoms of a molecular system.



One small step for a human, many small steps for a computer

Phase space sampling



Newton's equation of motion

acceleration = force /mass

$$\frac{d^2 r_i}{dt^2} = \frac{F_i}{m_i}$$

force = -slope of the potential

$$F_i = -\frac{\partial}{\partial r_i} V(r_1, r_2 \ldots\ldots\ldots r_N)$$

$$i = 1 \ldots\ldots \mathrm{N}_{\mathrm{particles}}$$

One small step for a human,
many small steps for a computer

Difference between methods

- Monte Carlo (MC) simulations
- Molecular Dynamics (MD) simulations

FIVE points to think about

- generation an ensemble of conformations
- information on time dependence of properties
- contributions to the total energy
- role of the potential energy
- factors that regulate the efficiency of method


KTH
VETENSKAP
OCH KONST

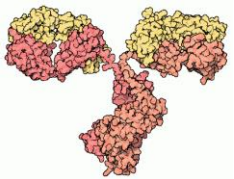
Challenge for biomolecular simulations

Biomolecules

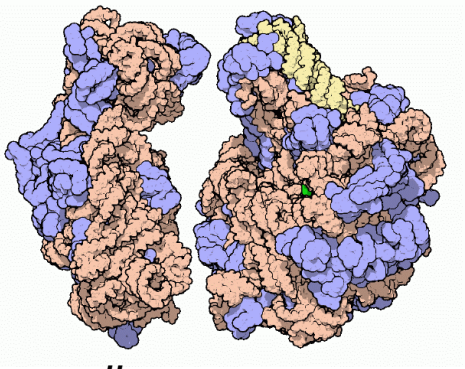
The four major types of biomolecules are carbohydrates, lipids, nucleic acids, and proteins.



water



antibody



ribosome

0.1 nm

30 nm

dimension (nanometer)

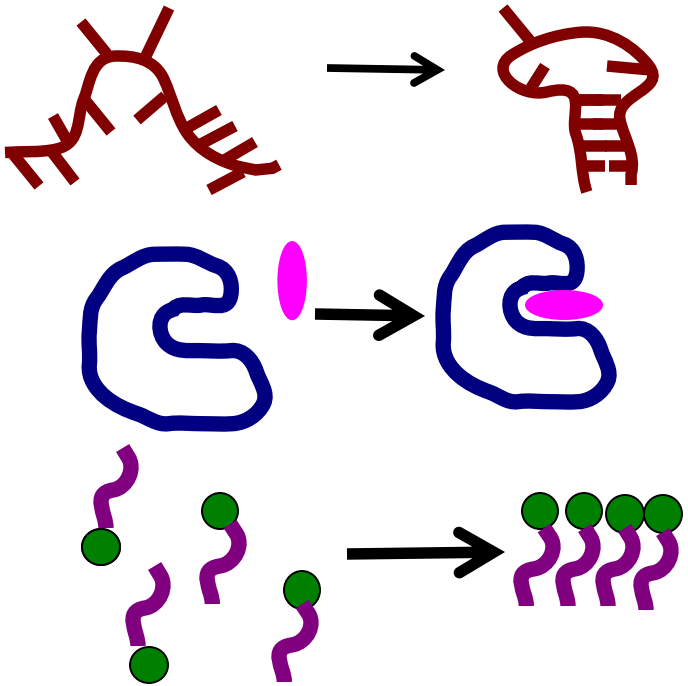
1 nm (10^{-9} m)

Many processes in the cell

Folding

Binding

Aggregation



driven by a tiny balance of several different force

What we need to describe a molecule

- a “model” that describes the molecular system and its degree of freedom
- **energy functions** or interaction potentials that describe the interactions between the particles in the molecular system used to calculate the potential energy of molecular system.

A simplify representation of a molecular system should be as simple as possible

- Why?

Have a look at the Nobel lectures in chemistry 2013
(Martin Karplus, Michael Levitt, Arieh Warshel)

Choose the appropriate molecular model

- Degree of freedom (which particles)
- Energy functions or interaction potentials that describe the interactions between the particles

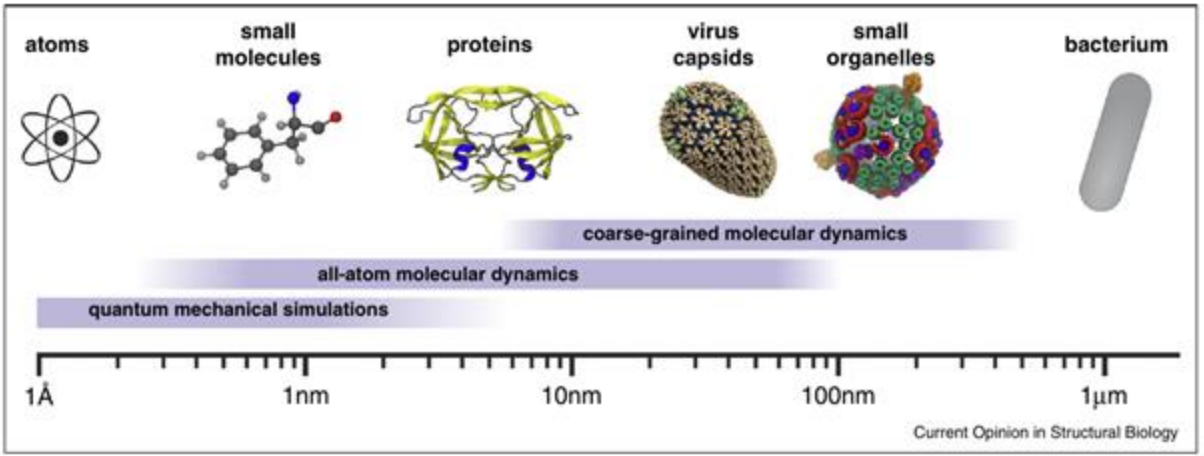
Reliability:

- Model must encompass the properties of interest

- Simulation time >> time scale of the process to be investigated

- Simulation size >> size of the simulated system

Dimension vs molecular models

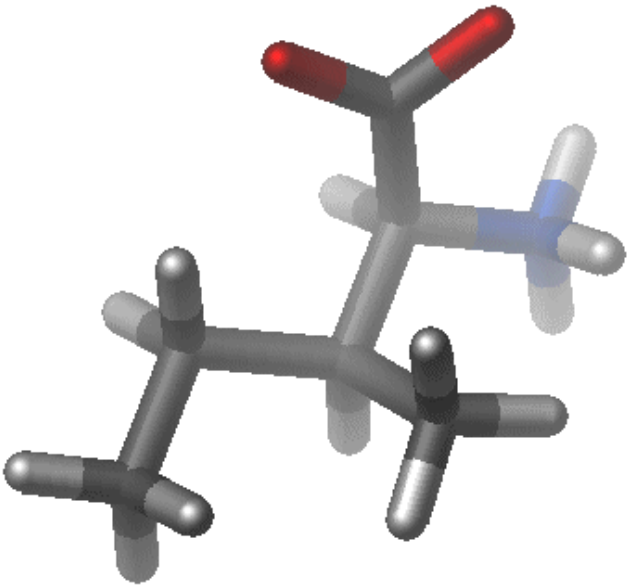


degree of freedom

Current Opinion in Structural Biology 2015, **31**:64–74

Pair approximation – effective potential

Molecular “movements” at atomistic level



Type of interactions

Potential energy

Potential energy can be divided into terms depending on the position of individual particles, pairs, triplets

more costly

$$V(r) = V(r_1, r_2, \cdots, r_N) =$$

$$\sum_i V^{(1)}(r_i) + \qquad \text{1-body}$$

$$\sum_{i,j} V^{(2)}(r_i, r_j) + \qquad \text{2-body}$$

$$\sum_{i,j,k} V^{(3)}(r_i, r_j, r_k) + \qquad \text{3-body}$$

$$\sum_{i,j,k,l} V^{(4)}(r_i, r_j, r_k, r_l) + \qquad \text{4-body}$$

$$\cdots + \sum_{i_1,\ldots,i_N} V^{(N)}(r_{1\mathrm{i}}, r_1, \cdots, r_N) \qquad \text{N-body}$$

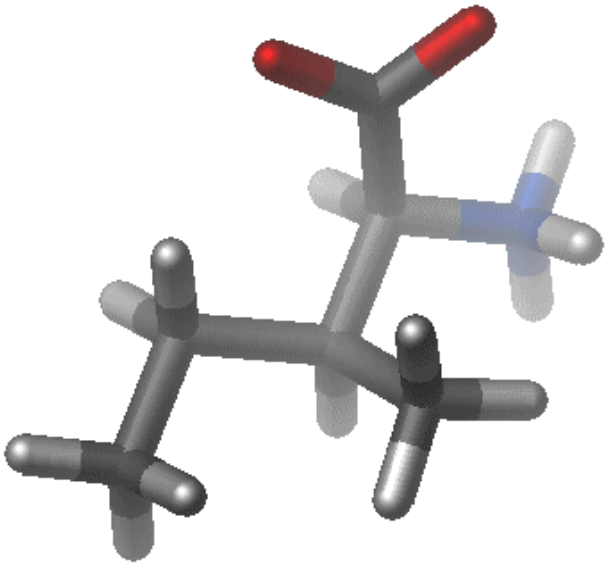
Interactions

Pair potential – effective potential

The pair approximation gives remarkably good description if the average of three-body effects can be partially included by defining an 'effective' pair potential

$$V(r) = V(r_1, r_2, \cdots, r_N)$$
$$= \sum_i V^{(1)}(r_1) + \sum_i Veff^{(2)}(r_1, r_2)$$

Consequence of this approximation: effective potential depends on density and temperature, while the true two-body potential does not.



Atomistic model

Interactions between particles

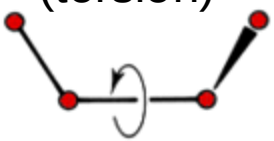
(bond)



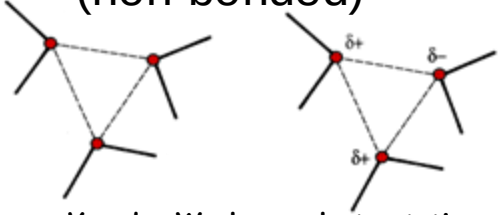
(angle)



(torsion)



(non-bonded)



Classical force field

Basic idea of molecular mechanics or force field methods:

- Ignore electronic motion and calculate the energy of a system as a function of nuclear position only (Born-Oppenheimer approximation)

- Particles of the system can be described as sphere and interactions with simple potential coming from classical mechanics

- Transferability: parameters developed for a system (e.i. small molecules) can used in other systems for similar chemical group (e.i. protein)

Atomistic model

Given a set of particles the force field is the potential function describing the inter-atomic interactions

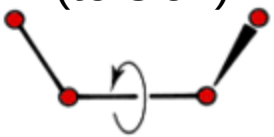
V = V(bond) **+** V(angle)





From Hooke

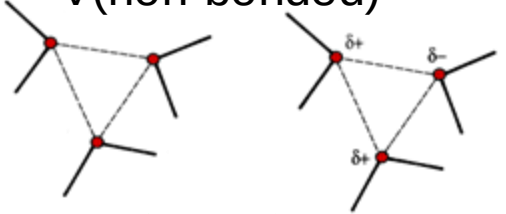
+ V(torsion)



From Fourier

+ V(other term)

+ V(non-bonded)



From Van der Waals

Electrostatic potential

Functional form in force field

Functional form is a compromise between accuracy and computational effort (the most accurate functional form may often be unsatisfactory)

It is desirable to be able to calculate the first derivatives of the energy, e.i. in order to use Molecular Dynamics simulation

Empirical force fields

$$V(r_1, r_2, ..., r_N) = \sum_{bonds} \frac{1}{2} K_b (b - b_0)^2 + \sum_{angles} \frac{1}{2} K_\theta (\theta - \theta_0)^2$$

$$+ \sum_{torsions} \frac{1}{2} K_\xi (\xi - \xi_0)^2$$

$$+ \sum_{torsions} \frac{1}{2} K_\phi [1 + \cos(n\phi + \delta)]$$

bonded-interactions

$$+ \sum_{pairs} [C_{12}(i,j) / r_{i,j}^{12} - C_6(i,j) / r_{i,j}^{6}] + q_i q_j / 4\pi\varepsilon_0 \varepsilon r_{i,j}$$

non-bonded interactions

Every force field has his own analytical functions and set of parameters

Parameters in force fields

- From experimental and *ab initio* studies of small model compounds
- Crystal data (bond lengths, angles)
- Spectroscopy (eg, IR, Raman) can give force constants for bonds, angles
- Charges usually by comparison with QM electrostatic potential;
- Values corresponding to observed data are computed with trial energy function and parameters are varied until a reasonable fit is obtained
- One hopes then that these parameters can be transferred to the large molecules of interest

Force fields and parameters

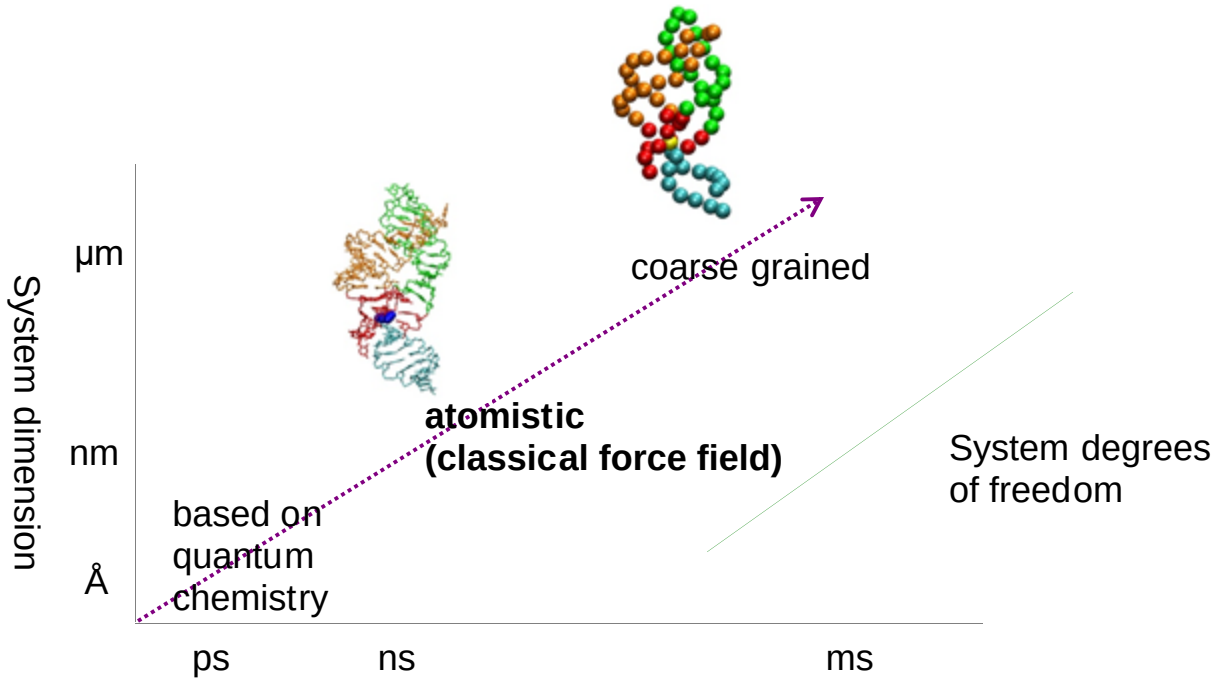
- Usually based on atom types, often many for each element
- Parameters intimately co-dependent on functional forms and each other
- Do not expect to mix and match force field parameters
- Addition of new parameters in line with parameterization strategy

- Most common biomolecular force field families: AMBER, CHARMM, GROMOS, OPLS, Martini..
- Generalized force field
- Open force field initiative (https://openforcefield.org/)

- AI-based force field
- Polarizable force field

- Online servers / offline tools where possible (SwissParam, ATB, Antechamber, LEaP, acpype, PRODRG, STaGE, MKTOP, CHARMM-GUI)

Molecular models



Dimension/time and molecular models

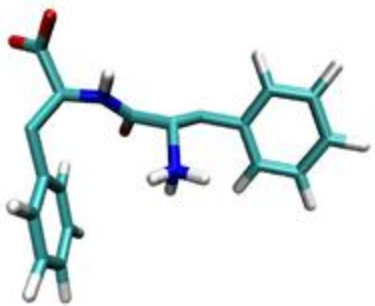
Atomistic molecular simulation - length

RESOLUTION	ATOMIC (10^{-10} meter)
SYSTEM SIZE	10-100 nm (10^{-7}-10^{-8} meter)

Atomistic molecular simulation – time
(only for MD simulation)

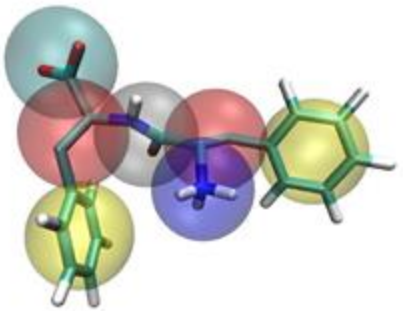
RESOLUTION	10^{-15} second (fs)
LENGTH TIME	10^{-6} second (μs)

Atomistic vs coarse-grained model



Degree of freedom reduction

Back-mapping to ato



ATOMISTIC
40 particles

COARSE GRAINED
7 particles

a fragment and structural based coarse grained approach
(1 up to 6 bead for residue)

Villa et al. PCCP 2009

What governs a molecular simulation

- choice of degrees of freedom
- force field parameters
- boundary conditions
- treatment of non-bonded interactions
- integration time step
- treatment of temperature and pressure
- environment (solvation effects, ions and etc)
- starting configuration

Molecular models

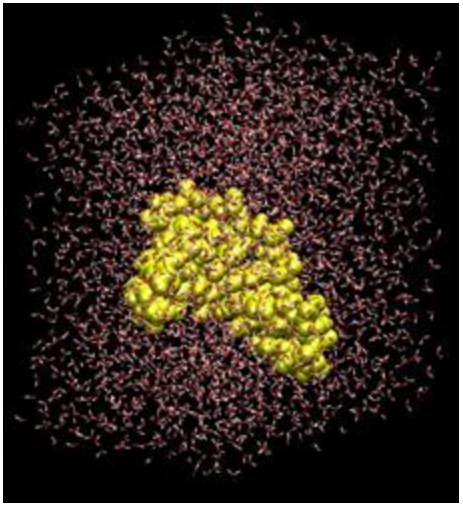
One of the factors that governs a MD simulation

What governs a molecular simulation

Boundary Conditions



Molecule in a box



What governs a molecular simulation

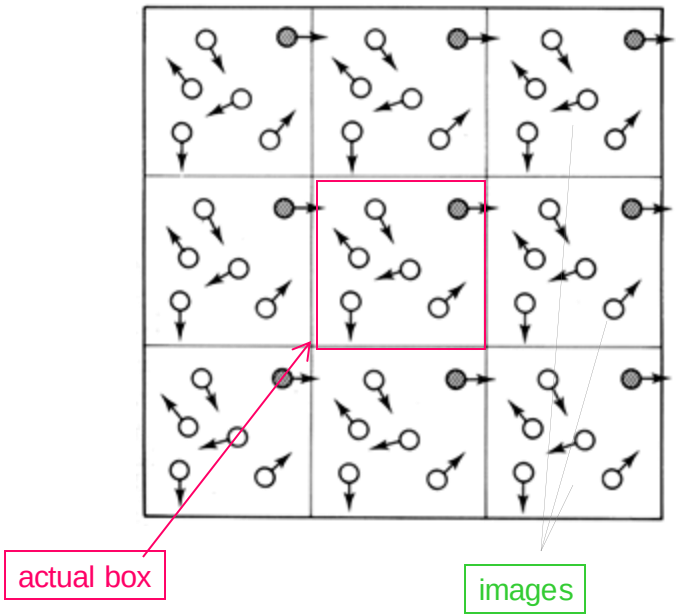
• Periodic boundaries

•Simulation box is replicated in all directions

•Particles leaving the central box are reintroduced on the opposite side

•Interactions calculated with particles in neighboring cells

•No vacuum interface anymore!



The actual box is surrounded by **26** images of its-self

Integration time step

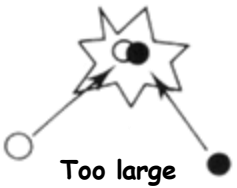
The time step determines how much time can be simulated.
The smaller the time step the more expensive the calculation

Choosing the time step

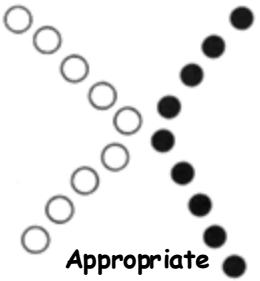
•What is an appropriate time step for the integration?



Too small
nothing happens



Too large
instabilities



Appropriate

System	Types of motion	Suggested time step
Rigid molecules	translation, rotation	5 fs
Flexible molecules, rigid bonds	translation, rotation, torsions	2 fs
Flexible molecules, flexible bonds	Translation, rotation, torsions, vibrations	1 or 0.5 fs

Integration time step

Kinetic energy = $1/2\ mv^2$

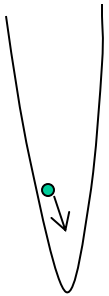
From equipartition theorem $\Rightarrow$ kinetic energy = $1/2k_BT$
(per degree for freedom)

mass of atom $\downarrow$ velocity of atom $\uparrow$ need smaller timestep

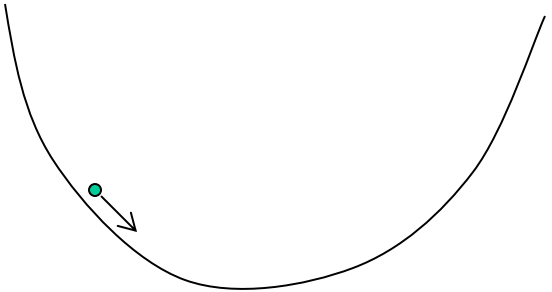
temperature $\uparrow$ velocity of atom $\uparrow$ need smaller timestep

Time step determined by lightest atom

The maximum time step determined by the curvature of the potential



Steep potential small time step



Shallow potential large time step

Temperature and Pressure

Thermodynamic ensembles:

- ***NVE* ensemble**

number of particles *N*, volume *V* and total energy E_{tot} of the system are conserved (Energy conservation)

$$E_{tot} = V_{potential} + K_{kinetic}$$

Rarely used in the biophysical and biochemical experiment

Thermodynamic ensembles:

- ***NVT* ensemble**

number of particles *N*, volume *V* and temperature T of the system are constant

- ***NPT* ensemble**

number of particles *N*, pressure p and temperature T of the system are constant

Experiments are often performed at constant pressure *P* and temperature *T.* Approximations (cut-offs, integration) can lead to instabilities and energy and/or temperature variations

Temperature

Temperature *T* related to the kinetic energy *K* of the system and therefore to the velocities

$$E_{kin} = \sum_{i=1}^{N} m_i \frac{v_i^2}{2} = \frac{N_{df} k_B T}{2}$$

N_{df} is the number of degrees of freedom in the system,

The temperature can thus be controlled by modifying the velocities of particles in the system

Pressure

The pressure is related to the volume of the system and the interaction between particles

Ideal gas: $PV = Nk_bT$

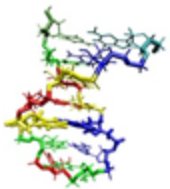
Real system: $PV = Nk_bT - \frac{1}{3k_bT} \sum_{i=1}^{N} \sum_{j=i+1}^{N} \vec{r}_{ij}\vec{f}_{ij}$

The second term is the virial describing the contribution due to the forces between the particles and it can easily be calculated in MD when calculating the forces.

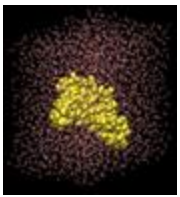
Pressure can be controlled by varying the volume of the system and scaling the position of the molecules

Molecular simulation

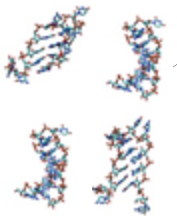
Starting structure



Conditions



OUTPUT:
Ensemble of structures (called configurations)

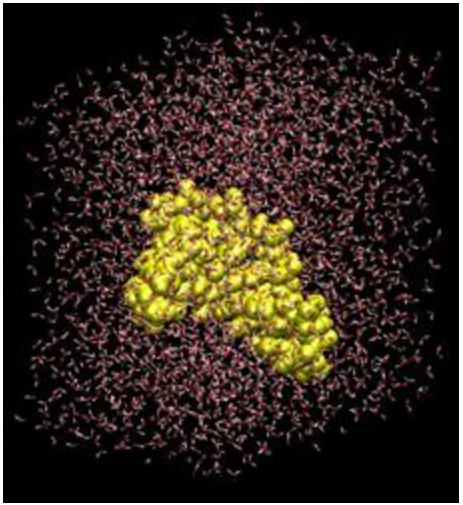


properties

Potential function to describe the molecular system

Simulation methods:
- ✓ Monte Carlo method
- ✓ Molecular dynamics method

How to start a simulation?



Molecular simulations

Set of initial conditions (starting structure)

for a Monte Carlo simulation

- N x, y, z coordinates (3N coordinates) describing the position of N atoms in the system.

for a Molecular Dynamics simulation

- N x, y, z coordinates describing the position of atoms in the system
- N x, y, z velocities describing the velocities of atoms in the system a t=0

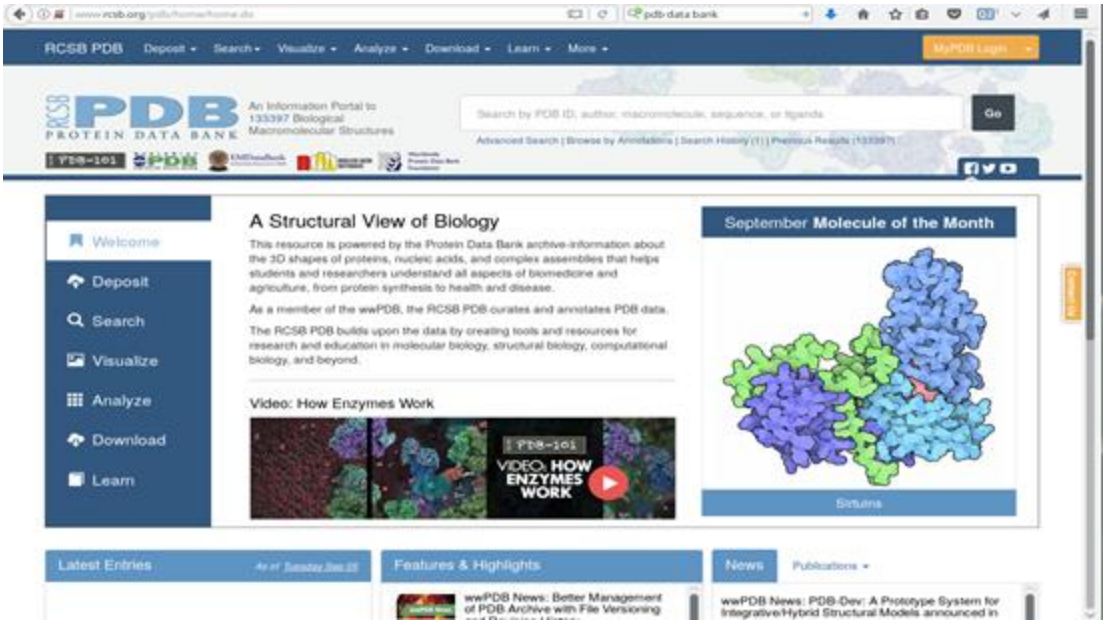
Molecular simulations

Initial coordinates

- Experimental structures from database (coming from X-ray crystallography, NMR spectroscopy, cryo-EM)
- Self-built model three-dimensional structure (e.i homology modelling)
- Pre-built solvent box

The structure at time t=0, r(0) for biomolecules are usually taken from experimental data (if available)

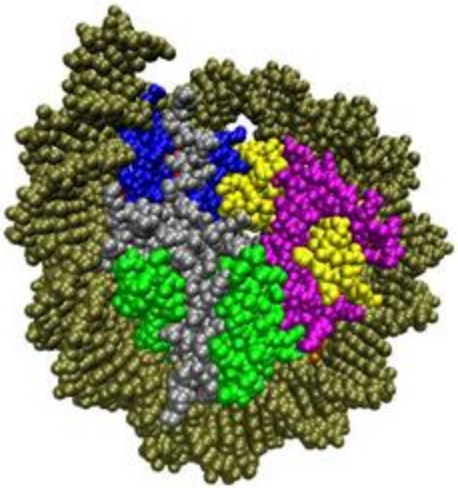
3-D structures Database



Search by PDB ID, author, biomolecule name, ligands, sequence

http://www.rcsb.org/

Three dimensional (3D) structure



Some visualization softwares:
VMD http://www.ks.uiuc.edu/Research/vmd/
Rasmol http://www.openrasmol.org/
Pymol http://www.pymol.org/

Molecular simulations

Initial velocities

The velocities at time t=0 $v_i(0) = \frac{dr_i}{dt}$, $i = 1 \ldots\ldots\ldots \mathrm{N}_{\mathrm{particles}}$

Usually the velocities are taken from random (Maxwell) distribution with the kinetic energy corresponding to desired temperature (T)

$$E_{kin} = \sum_{i=1}^{N} m_i \frac{v_i^2}{2} = \frac{N_{df} k_B T}{2} = \frac{(3N-6) k_B T}{2}$$

Software for molecular dynamics

AMBER www.ambermd.org
CHARMM www.charmm.org
NAMD http://www.ks.uiuc.edu/Research/namd/
GROMOS www.igc.ethz.ch/gromos

GROMACS www.gromacs.org

community-driven project, built on contributions from all over the world. At present, the development is being led by researchers from the KTH Royal Institute of Technology

LAMMPS www.lammps.org

Molecular Dynamics Simulations



www.gromacs.org

What is the name of the bird in GROMACS logo?

molecular simulation – is usu*ally associated with computer modelling*

Which machine you will use it depends on the molecular system and on the question to answer



~ 4-8 CPU nodes
~ 1 GPU

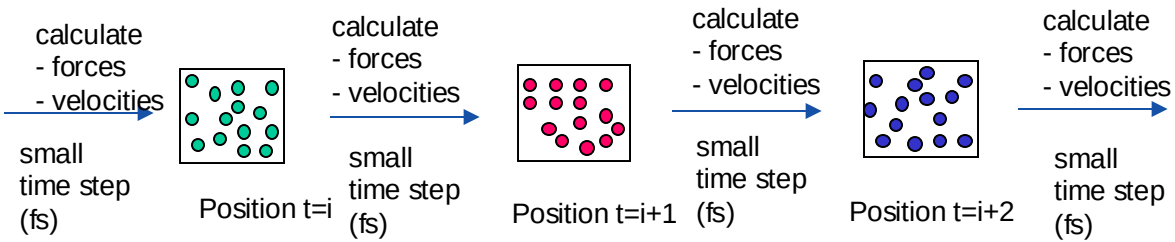


~ 554 CPU nodes
~ 56 GPU nodes

Why good performance

One goal of a molecular dynamics simulation is to generate enough representative conformations of the molecular system in such a way that accurate values of a property can be obtained.

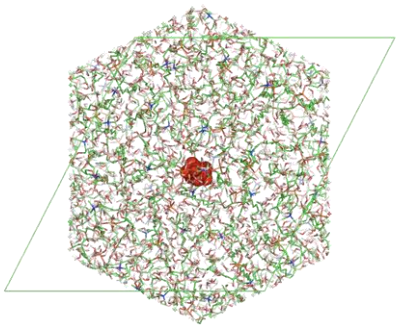
How? by iteratively solving equations of motion



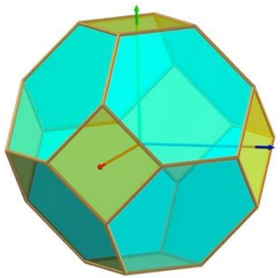
Some interactions take more time than other

Addressing system size

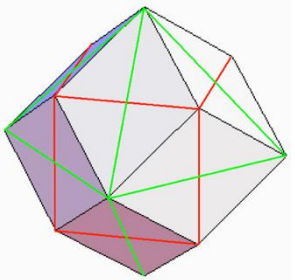
- Cubic / rectangular
- Hexagonal for membrane simulations (volume 87% of cube)
- Truncated octahedron (approximate sphere, volume 77% of a cube)
- Rhombic dodecahedron (most spherical cell, 71%)



Hexagonal



Truncated octahedron



Rhombic dodecahedron

Addressing the time-step

Δt limited by fast motions - 1fs

=> Remove bond vibrations => 2 fs

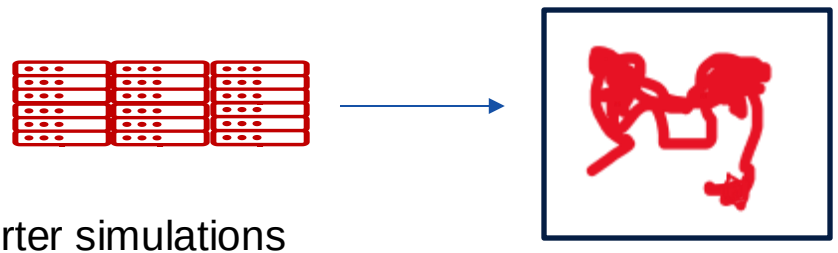
- Constraint algorithms (LINCS, P-LINCS, SHAKE)

- Multi-time step => different forces are evaluated at different time steps
- Mass-repartitioning

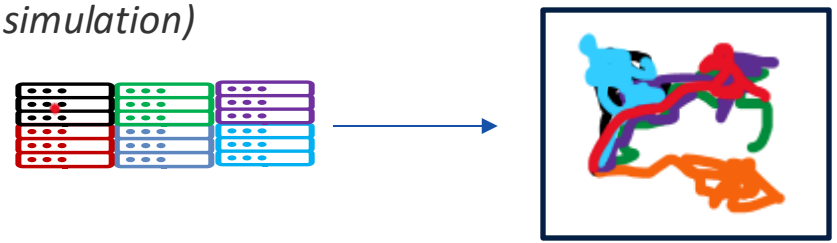
Addressing sampling & architecture needs

to sample the conformational space.

- One long simulation



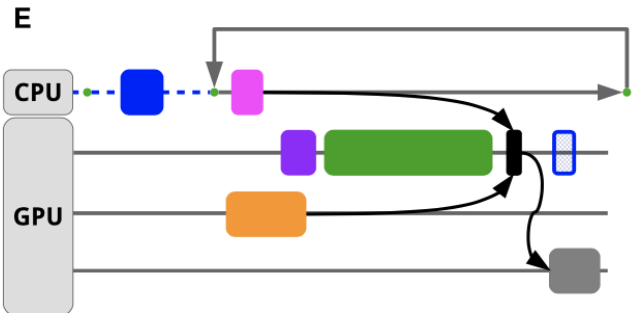
- Many shorter simulations
- (*ensemble simulation)*



Mapping software and hardware
What on CPU and GPU?

GROMACS

CPU is used for scheduling work, transferring data, and launching computation on the accelerator, as well as inter- and intra-node communication.
Accelerator tasks are launched asynchronously using APIs to allow concurrent CPU–GPU execution

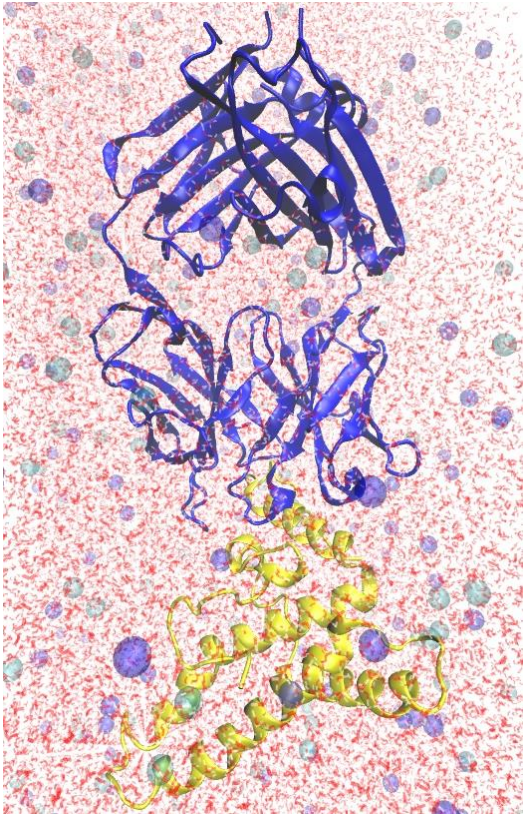


Non bonded forces
PME forces
Bonded forces

dynamic list pruning
integration, constrains

J. Chem. Phys.. 2020;153(13). doi:10.1063/5.0018516

Time - Reference



Antigen-Antibody system

- 150 000 atoms

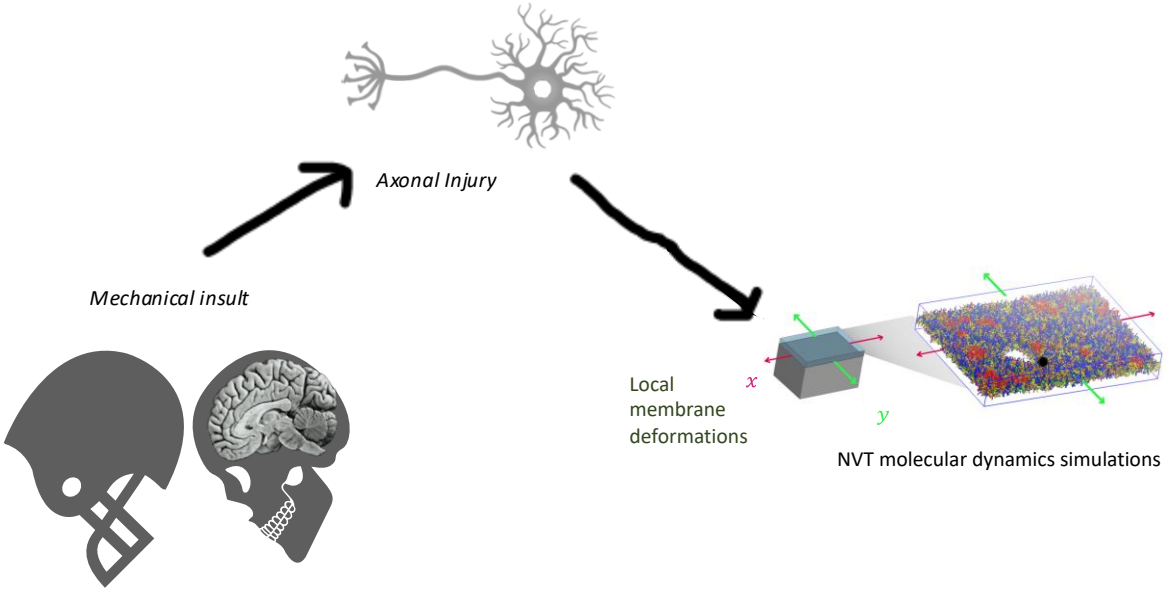
- on 64 core on Dardel supercomputer

- **software GROMACS**
- **1.5 ms/step wall time**

Small systems enable more & longer sampling

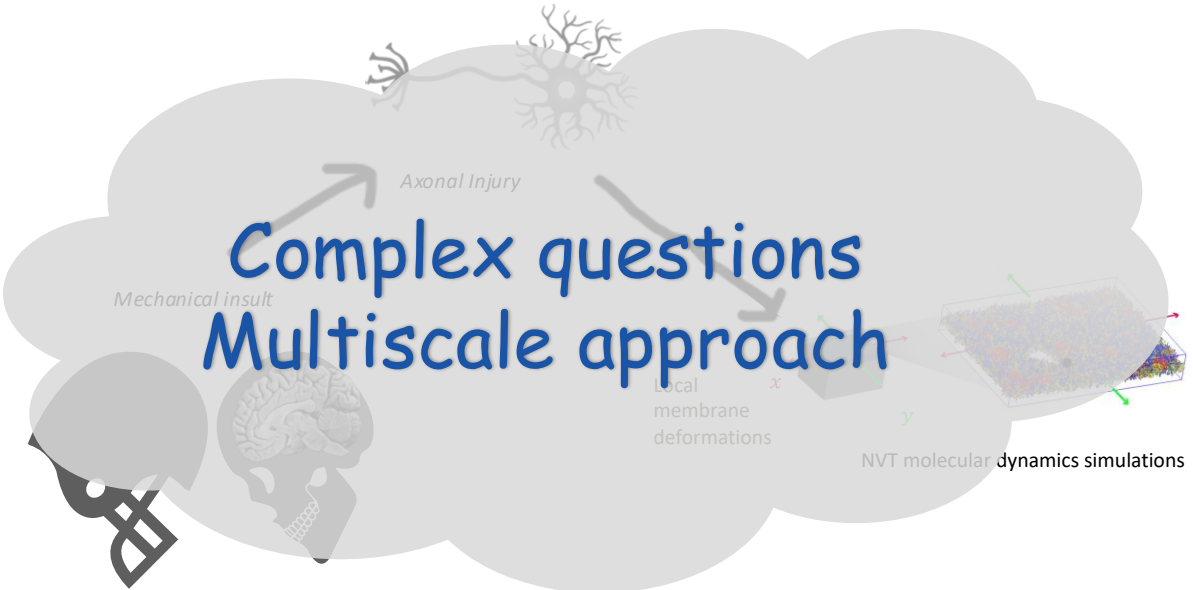
1μs/day: Understanding motions	432 μs/step walltime
10μs/day: Predicting motions Interpreting/improving experiments	43 μs/step walltime
>100μs/day: Replacing biochemistry experiments	4 μs/step walltime
>1ms/day: Replacing medicine/biology experiments	400 ns/step walltime

Understand injuries via reducing a complex problems to simpler systems



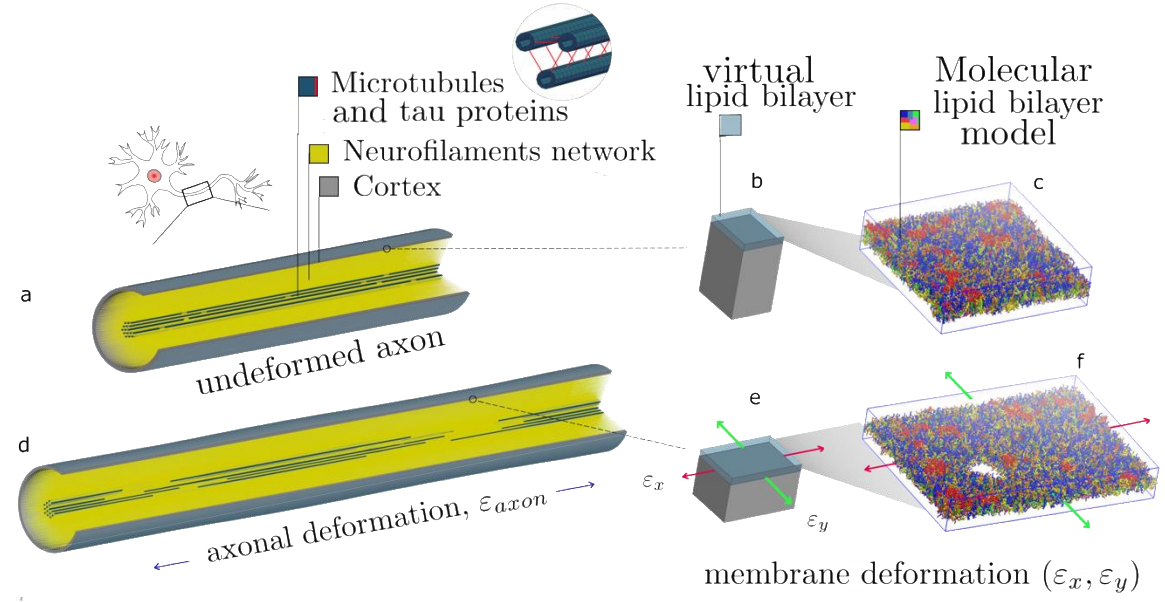
A Montanino et al. "Localized axolemma deformations suggest mechanoporation as axonal injury trigger" *- Frontiers in neurology, 2020*

Understand injuries via reducing a complex problems to simpler systems



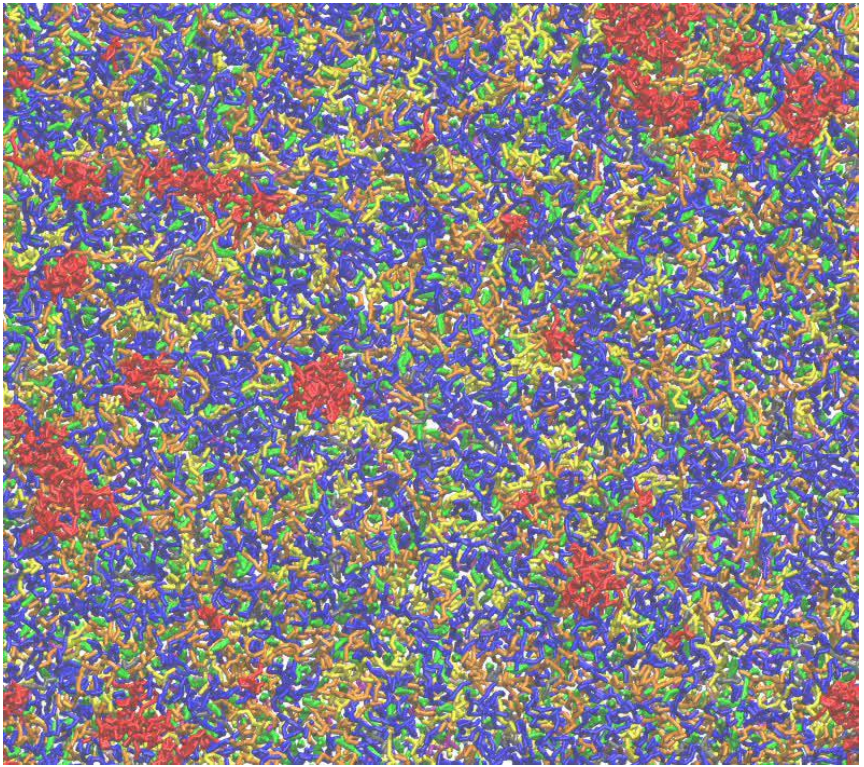
A Montanino et al. "Localized axolemma deformations suggest mechanoporation as axonal injury trigger" *- Frontiers in neurology, 2020*

Finite element/Molecular models



Montanino et al. "Localized axolemma deformations suggest mechanoporation as axonal injury trigger" (10.3389/fneur.2020.00025)

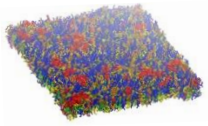
Spontaneous pore formation



From axon deformation to local deformation

ε_{axon}

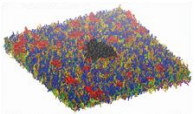
Membrane model



ε_{axon}	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
$\dot{\varepsilon} = 1s^{-1}$	0	0.02	0.05*	0.07*	0.10*	0.13*	0.17*	0.20*	0.24*	0.27*	INTD 0.31*	0.34*	0.37	0.40*	0.44	0.47*
$\dot{\varepsilon} = 10s^{-1}$	0	0.02	0.05*	0.07*	0.10*	0.13*	0.16	0.20*	0.26	INTD 0.32*	0.38*	0.44	0.51	0.57	0.63	0.69
$\dot{\varepsilon} = 40s^{-1}$	0	0.02	0.04	0.05*	0.07	0.09	0.11	0.14	0.16	0.19	0.22*	INTD 0.28*	0.36*	0.43*	0.50	0.5

ΔP_m min — max

Membrane-protein model



ε_{axon}	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
$\dot{\varepsilon} = 1s^{-1}$	0	0.02	0.05*	0.07*	0.10	0.13	0.17*	0.20	0.24	0.27	0.31	0.34*	0.37	0.40	INTD 0.44*	0.47*
$\dot{\varepsilon} = 10s^{-1}$	0	0.02	0.05*	0.07*	0.10	0.13	0.16	0.20	0.26	0.32	0.38	INTD 0.44*	0.51*	0.57	0.63	0.69
$\dot{\varepsilon} = 40s^{-1}$	0	0.02	0.04	0.05*	0.07*	0.09	0.11	0.14	0.16	0.19	0.22	0.28*	0.36	INTD 0.43*	0.50	0.5

ΔP_m min — max

poration

interdigitation INTD

Do you want to try to perform a molecular dynamics simulation of protein using GROMACS?

Go to
https://tutorials.gromacs.org/md-intro-tutorial.html

and follow the instruction to run online



THANK YOU !